

# Mobile Shop

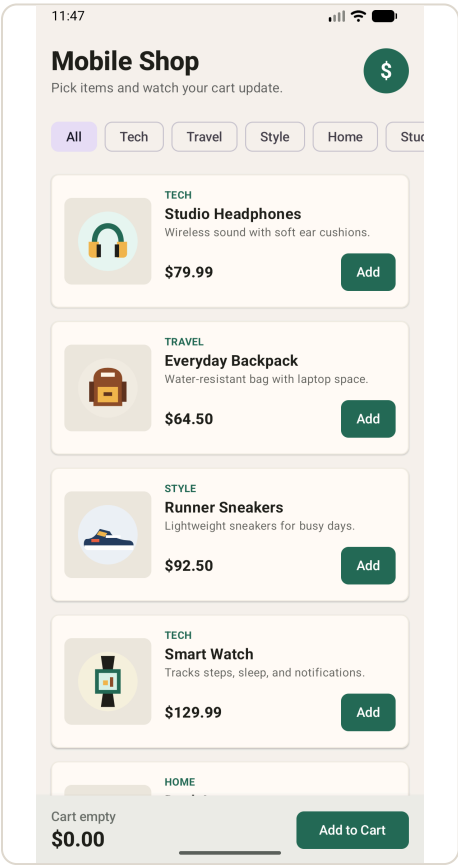
Mobile Shop is a Kotlin Jetpack Compose Android shopping app. It shows a small product catalog, lets users filter by category, add products to a cart, review cart totals, and complete a secure demo checkout using a simulated payment method.

The app is intentionally local-only: product data, cart quantities, and simulated payment state live in memory. It does not collect or store real card numbers, passwords, tokens, or personal payment details.

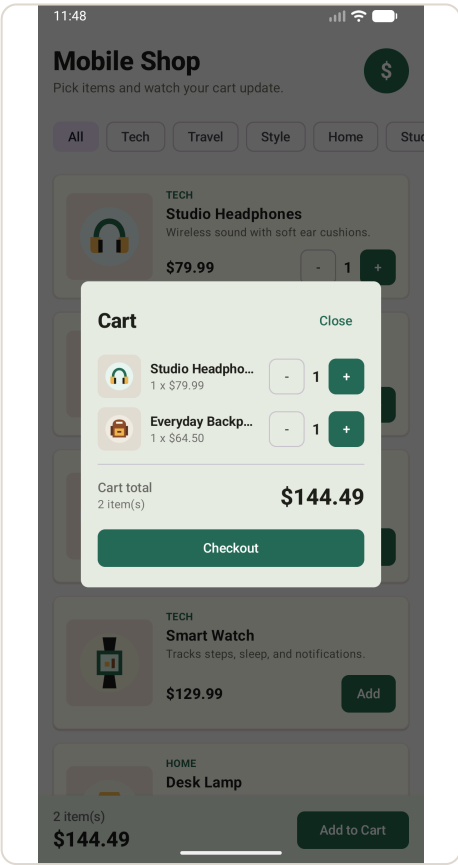
## Preview

GitHub will render these screenshots directly from the docs/screenshots folder.

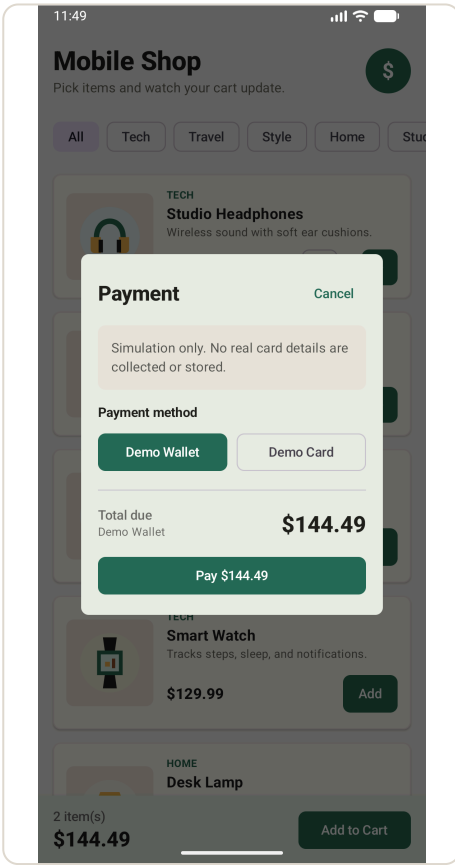
Product Catalog



Cart



Simulated Payment



## Features

- Kotlin + Jetpack Compose Android app.
- In-memory product catalog with six products and custom vector drawable artwork.
- Category filter chips for All, Tech, Travel, Style, Home, and Study.
- Quantity controls for adding and reducing cart items.
- Cart summary with live item count and total price.
- Mobile cart dialog opened from the bottom Add to Cart bar.
- Wide-screen layout with catalog and cart displayed side by side.
- Checkout flow with a simulated payment dialog.
- Demo payment options: Demo Wallet and Demo Card.
- Secure demo payment behavior: no real payment details are requested, saved, or transmitted.

- Android hardening: app backup disabled, cleartext traffic disabled, release minification enabled, and release resource shrinking enabled.
- Unit tests for cart totals, item counts, and currency formatting.

## Configure and Run

1. Install Android Studio with Android SDK Platform 36.
2. Use Java 21 for Gradle. This project pins Gradle to `/opt/homebrew/opt/openjdk@21/libexec/openjdk.jdk/Contents/Home` in `gradle.properties`.
3. Open the project root folder, the folder containing `settings.gradle.kts`, in Android Studio.
4. Let Android Studio sync Gradle.
5. Start an Android emulator or connect an Android phone.
6. Press Run in Android Studio, or use the command line below.

```
./gradlew test assembleDebug
adb install -r app/build/outputs/apk/debug/app-debug.apk
adb shell am start -n com.example.mobileshop/.MainActivity
```

The debug APK is generated at `app/build/outputs/apk/debug/app-debug.apk`.

To create the hardened release build:

```
./gradlew assembleRelease
```

The release APK is generated at `app/build/outputs/apk/release/app-release-unsigned.apk`.

## Project Structure

```
MobileShop/
|-- app/
|   |-- build.gradle.kts
|   |-- proguard-rules.pro
|   |-- src/
|       |-- main/
|           |-- AndroidManifest.xml
|           |-- java/com/example/mobileshop/
|               |-- MainActivity.kt
|               |-- ShopModels.kt
|           |-- res/
|               |-- drawable/
|                   |-- ic_store_bag.xml
|                   |-- product_backpack.xml
|                   |-- product_headphones.xml
|                   |-- product_lamp.xml
|                   |-- product_notebook.xml
|                   |-- product_sneakers.xml
|                   |-- product_watch.xml
|               |-- values/
|                   |-- colors.xml
|                   |-- strings.xml
|                   |-- themes.xml
|           |-- test/java/com/example/mobileshop/
|               |-- ShopModelsTest.kt
|-- docs/screenshots/
|   |-- bag.png
|   |-- cart.png
|   |-- catalog.png
|   |-- payment.png
```

```
|-- gradle/wrapper/  
|   |-- gradle-wrapper.jar  
|   `-- gradle-wrapper.properties  
|-- build.gradle.kts  
|-- gradle.properties  
|-- gradlew  
|-- gradlew.bat  
|-- README.md  
|-- README.html  
|-- README.pdf  
`-- settings.gradle.kts
```

## App Flow

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1. Browse products in the catalog.
2. Use category chips to filter visible products.
3. Tap Add on a product, then use + and – to adjust quantity.
4. On phones, tap the bottom Add to Cart button to open the cart.
5. Review items and the cart total.
6. Tap Checkout.
7. Choose Demo Wallet or Demo Card.
8. Tap Pay to complete the simulated payment and clear the cart.

## Security Notes

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This project uses a simulated payment flow for coursework/demo purposes. It is safer than collecting fake card numbers because the UI never asks for sensitive card data.

Current hardening settings:

- android:allowBackup="false"
- android:fullBackupContent="false"
- android:usesCleartextTraffic="false"
- isMinifyEnabled = true for release builds
- isShrinkResources = true for release builds

There is no backend, authentication system, database, network API, or real payment processor in this app.

## Verification

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The commands below were run successfully after the latest updates.

```
./gradlew :app:assembleDebug  
./gradlew :app:assembleRelease
```

Result: **BUILD SUCCESSFUL.**